05. Schedule Evaluation

Objective and Scope

The objective of this section is to evaluate the schedule performance of the Sunset Ridge Apartments project, using the baseline schedule and actual progress data. Specifically, it aims to identify critical path activities, assess schedule variances, and provide a data-driven foundation for forecasting project completion and supporting proactive schedule management.

Key analysis includes:

- Critical path identification and dependency mapping
- Schedule variance analysis (planned vs. actual start and finish dates)
- Activity-level performance review
- Identification of recurring delays and bottlenecks
- Preparation of structured inputs for forecasting and scenario modeling

Source: Sunset Ridge Apartments Baseline Schedule (Primavera P6 export), Weekly Progress Reports, Subcontractor Daily Logs, Inspection Records

Data Sources and Preparation

Primary data sources:

- Baseline schedule: Sunset_Ridge_Baseline_Schedule.xlsx
- Actual progress: Weekly_Progress_Log.xlsx, Subcontractor_Daily_Log.xlsx, Inspection_Log.xlsx

Data preparation steps:

- Standardized activity IDs and dates across baseline and actual datasets
- Calculated planned and actual durations for each activity
- Flagged missing actual dates for in-progress activities
- Joined baseline and actual data on activity_id for variance analysis

Source: Sunset_Ridge_Baseline_Schedule.xlsx, Weekly_Progress_Log.xlsx, Subcontractor_Daily_Log.xlsx, Inspection_Log.xlsx

Critical Path Identification

The critical path was identified using the baseline schedule. Activities on the critical path have zero total float and directly impact the project completion date if delayed.

Critical path activities:

- A1000 Site Mobilization
- A1010 Foundation Excavation
- A1020 Foundation Pour
- A1030 Structural Steel Erection
- A1040 Roof Framing
- A1050 Roof Sheathing
- A1060 Roof Shingles
- A1070 Exterior Finishes
- A1080 Final Inspection

Source: Sunset_Ridge_Baseline_Schedule.xlsx

Critical Path Activities Schedule Variance

Activity	Planned Start	Actual Start	Start Variance (days)	Planned Finish	Actual Finish	Finish Variance (days)
A1000 Site Mobilization	2025-03-01	2025-03-01	0	2025-03-05	2025-03-05	0
A1010 Foundation Excavation	2025-03-06	2025-03-08	+2	2025-03-14	2025-03-16	+2
A1020 Foundation Pour	2025-03-15	2025-03-17	+2	2025-03-21	2025-03-23	+2
A1030 Structural Steel Erection	2025-03-22	2025-03-24	+2	2025-04-04	2025-04-07	+3
A1040 Roof Framing	2025-04-05	2025-04-08	+3	2025-04-18	2025-04-22	+4
A1050 Roof Sheathing	2025-04-19	2025-04-23	+4	2025-04-25	2025-04-29	+4
A1060 Roof Shingles	2025-04-26	2025-04-30	+4	2025-05-02	2025-05-06	+4
A1070 Exterior Finishes	2025-05-03	2025-05-07	+4	2025-05-16	2025-05-21	+5
A1080 Final Inspection	2025-05-17	2025-05-22	+5	2025-05-18	2025-05-23	+5

Source: Sunset_Ridge_Baseline_Schedule.xlsx, Weekly_Progress_Log.xlsx

Critical Path Delay Analysis

Total delay on critical path activities: 29 days

Delay breakdown by activity:

- A1010 Foundation Excavation: +2 days
- A1020 Foundation Pour: +2 days
- A1030 Structural Steel Erection: +3 days
- A1040 Roof Framing: +4 days
- A1050 Roof Sheathing: +4 days
- A1060 Roof Shingles: +4 days
- A1070 Exterior Finishes: +5 days
- A1080 Final Inspection: +5 days

Source: Sunset_Ridge_Baseline_Schedule.xlsx, Weekly_Progress_Log.xlsx

Delay Root Cause Analysis

Root cause analysis was performed using the 5 Whys method for the two activities with the highest delay: A1030 Structural Steel Erection (+3 days) and A1040 Roof Framing (+4 days).

A1030 Structural Steel Erection (+3 days)

- Why 1: Steel delivery was delayed by 2 days.
- Why 2: Supplier had a production backlog due to high demand.
- Why 3: Supplier was not notified of the schedule change in advance.
- Why 4: Procurement team did not have a formal notification process for schedule changes.
- Why 5: No process was defined for communicating schedule changes to suppliers.

A1040 Roof Framing (+4 days)

- Why 1: Roof framing was delayed by 4 days.
- Why 2: Crew was reassigned to other tasks while waiting for steel.
- Why 3: Crew was not informed of the steel delay in time to plan other work.
- Why 4: Weekly progress meeting was not used to communicate cross-trade dependencies.
- Why 5: No cross-trade coordination process was in place.

Source: Weekly_Progress_Log.xlsx, Subcontractor_Daily_Log.xlsx

Recurring Delay Patterns

Recurring delay patterns observed across the project:

- Steel delivery delays affecting downstream trades
- Crew reassignment without formal communication
- Weekly meeting not used for cross-trade coordination
- Inspection scheduling not aligned with crew availability

Source: Weekly_Progress_Log.xlsx, Subcontractor_Daily_Log.xlsx, Inspection_Log.xlsx

Schedule Performance Summary

- Total activities: 8
- Activities completed: 8
- Activities on time: 0
- Activities delayed: 8
- Average finish variance: +4.0 days
- Total project delay: 29 days

Source: Sunset_Ridge_Baseline_Schedule.xlsx, Weekly_Progress_Log.xlsx

Recommendations

- Implement a formal supplier notification process for schedule changes.
- Use weekly progress meetings to coordinate cross-trade dependencies.
- Monitor steel delivery dates closely and escalate early if delays are expected.
- Update the schedule weekly to reflect actual progress and forecast impacts.

Source: Sunset_Ridge_Baseline_Schedule.xlsx, Weekly_Progress_Log.xlsx, Subcontractor_Daily_Log.xlsx, Inspection_Log.xlsx

Schedule Performance Dashboard

Source: Sunset_Ridge_Baseline_Schedule.xlsx, Weekly_Progress_Log.xlsx

06. Cost Analysis

Objective and Scope

The objective of this section is to evaluate the cost performance of the Sunset Ridge Apartments project, using the baseline budget and actual cost data. Specifically, it aims to identify cost variances, assess budget performance by activity and cost category, and provide a data-driven foundation for forecasting project cost and supporting proactive cost management.

Key analysis includes:

- Cost variance analysis (planned vs. actual costs)
- Activity-level cost performance review
- Identification of cost overruns and underruns
- Preparation of structured inputs for forecasting and scenario modeling

Source: Sunset Ridge Apartments Baseline Budget, Weekly Progress Reports, Subcontractor Daily Logs, Inspection Records

Data Sources and Preparation

Primary data sources:

- Baseline budget: Sunset_Ridge_Baseline_Budget.xlsx
- Actual cost: Weekly_Progress_Log.xlsx, Subcontractor_Daily_Log.xlsx, Inspection_Log.xlsx

Data preparation steps:

- Standardized activity IDs across baseline and actual datasets
- Calculated planned and actual costs for each activity
- Flagged missing actual costs for in-progress activities
- Joined baseline and actual data on activity_id for variance analysis

Source: Sunset_Ridge_Baseline_Budget.xlsx, Weekly_Progress_Log.xlsx, Subcontractor_Daily_Log.xlsx, Inspection_Log.xlsx

Cost Variance Analysis

Activity	Planned Cost	Actual Cost	Cost Variance	Variance %
A1000 Site Mobilization	12,000	12,000	0	0%
A1010 Foundation Excavation	18,000	19,500	+1,500	+8.3%
A1020 Foundation Pour	22,000	23,800	+1,800	+8.2%
A1030 Structural Steel Erection	35,000	38,500	+3,500	+10.0%
A1040 Roof Framing	28,000	31,200	+3,200	+11.4%
A1050 Roof Sheathing	15,000	16,500	+1,500	+10.0%
A1060 Roof Shingles	18,000	19,800	+1,800	+10.0%
A1070 Exterior Finishes	42,000	46,200	+4,200	+10.0%
A1080 Final Inspection	5,000	5,500	+500	+10.0%
Total	195,000	212,500	+17,500	+8.97%

Source: Sunset_Ridge_Baseline_Budget.xlsx, Weekly_Progress_Log.xlsx

Cost Variance Structure

Source: Sunset_Ridge_Baseline_Budget.xlsx, Weekly_Progress_Log.xlsx

Cost Drivers

Cost drivers observed across the project:

- Steel delivery delays affecting downstream trades
- Crew reassignment without formal communication
- Weekly meeting not used for cross-trade coordination
- Inspection scheduling not aligned with crew availability

Source: Weekly_Progress_Log.xlsx, Subcontractor_Daily_Log.xlsx, Inspection_Log.xlsx

Cost Performance Summary

- Total activities: 8
- Activities completed: 8
- Activities on time: 0
- Activities delayed: 8
- Average cost variance: +9.0%
- Total project cost variance: +8.97%

Source: Sunset_Ridge_Baseline_Budget.xlsx, Weekly_Progress_Log.xlsx

Recommendations

- Implement a formal supplier notification process for schedule changes.
- Use weekly progress meetings to coordinate cross-trade dependencies.
- Monitor steel delivery dates closely and escalate early if delays are expected.
- Update the schedule weekly to reflect actual progress and forecast impacts.

Source: Sunset_Ridge_Baseline_Budget.xlsx, Weekly_Progress_Log.xlsx, Subcontractor_Daily_Log.xlsx, Inspection_Log.xlsx

Cost Performance Dashboard

Source: Sunset_Ridge_Baseline_Budget.xlsx, Weekly_Progress_Log.xlsx

07. Forecasting

Objective and Scope

The objective of this section is to forecast the remaining cost and schedule performance of the Sunset Ridge Apartments project, using actual progress data and baseline performance indicators. Specifically, it aims to estimate the project's final cost and completion date, identify the most likely outcomes, and provide a data-driven foundation for scenario modeling and proactive management.

Key analysis includes:

- Cost forecasting using EAC (Estimate at Completion) and CPI (Cost Performance Index)
- Schedule forecasting using SPI (Schedule Performance Index)
- Identification of recurring delays and bottlenecks
- Preparation of structured inputs for scenario modeling

Source: Sunset Ridge Apartments Baseline Budget, Weekly Progress Reports, Subcontractor Daily Logs, Inspection Records

Data Sources and Preparation

Primary data sources:

- Baseline budget: Sunset_Ridge_Baseline_Budget.xlsx
- Actual cost: Weekly_Progress_Log.xlsx, Subcontractor_Daily_Log.xlsx, Inspection_Log.xlsx

Data preparation steps:

- Standardized activity IDs across baseline and actual datasets
- Calculated planned and actual costs for each activity
- Flagged missing actual costs for in-progress activities
- Joined baseline and actual data on activity_id for variance analysis

Source: Sunset_Ridge_Baseline_Budget.xlsx, Weekly_Progress_Log.xlsx, Subcontractor_Daily_Log.xlsx, Inspection_Log.xlsx

Cost Forecasting

Cost forecasting using EAC (Estimate at Completion) and CPI (Cost Performance Index):

- EAC = Actual Cost / CPI
- CPI = Planned Cost / Actual Cost

Activity	Planned Cost	Actual Cost	CPI	EAC
A1000 Site Mobilization	12,000	12,000	1.00	12,000
A1010 Foundation Excavation	18,000	19,500	0.92	21,000
A1020 Foundation Pour	22,000	23,800	0.92	25,800
A1030 Structural Steel Erection	35,000	38,500	0.91	42,300
A1040 Roof Framing	28,000	31,200	0.90	34,700
A1050 Roof Sheathing	15,000	16,500	0.91	18,100
A1060 Roof Shingles	18,000	19,800	0.91	21,800
A1070 Exterior Finishes	42,000	46,200	0.91	50,800
A1080 Final Inspection	5,000	5,500	0.91	6,000
Total	195,000	212,500	0.92	232,500

Source: Sunset_Ridge_Baseline_Budget.xlsx, Weekly_Progress_Log.xlsx

Source: Sunset_Ridge_Baseline_Budget.xlsx, Weekly_Progress_Log.xlsx

Schedule Forecasting

Schedule forecasting using SPI (Schedule Performance Index):

- SPI = Planned Duration / Actual Duration

Activity	Planned Duration	Actual Duration	SPI
A1000 Site Mobilization	5	5	1.00
A1010 Foundation Excavation	7	9	0.78
A1020 Foundation Pour	6	8	0.75
A1030 Structural Steel Erection	12	15	0.80
A1040 Roof Framing	14	18	0.78
A1050 Roof Sheathing	6	8	0.75
A1060 Roof Shingles	6	8	0.75
A1070 Exterior Finishes	10	13	0.77
A1080 Final Inspection	1	1	1.00
Average	7.5	9.8	0.77

Source: Sunset_Ridge_Baseline_Schedule.xlsx, Weekly_Progress_Log.xlsx

Source: Sunset_Ridge_Baseline_Schedule.xlsx, Weekly_Progress_Log.xlsx

Recommendations

- Implement a formal supplier notification process for schedule changes.
- Use weekly progress meetings to coordinate cross-trade dependencies.
- Monitor steel delivery dates closely and escalate early if delays are expected.
- Update the schedule weekly to reflect actual progress and forecast impacts.

Source: Sunset_Ridge_Baseline_Budget.xlsx, Weekly_Progress_Log.xlsx, Subcontractor_Daily_Log.xlsx, Inspection_Log.xlsx

Forecasting Dashboard

Source: Sunset_Ridge_Baseline_Budget.xlsx, Weekly_Progress_Log.xlsx

08. Scenario Modeling

Objective and Scope

The objective of this section is to model three scenarios for the Sunset Ridge Apartments project, using the forecasted cost and schedule performance from Section 07. Specifically, it aims to quantify the impact of recurring delays and bottlenecks, compare outcomes across scenarios, and provide a data-driven foundation for decision-making and proactive management.

Key analysis includes:

- Baseline scenario: current performance continues without intervention
- Mitigation scenario: recurring delays are reduced by 50%
- Worst case scenario: delays worsen by 25%
- Comparison of cost and schedule outcomes across scenarios

Source: Sunset Ridge Apartments Forecasted Cost and Schedule Performance, Weekly Progress Reports, Subcontractor Daily Logs, Inspection Records

Data Sources and Preparation

Primary data sources:

- Forecasted cost: Sunset_Ridge_Forecast.xlsx
- Forecasted schedule: Sunset_Ridge_Forecast.xlsx

Data preparation steps:

- Standardized activity IDs across forecast datasets
- Calculated baseline, mitigation, and worst case scenarios
- Joined forecast data on activity_id for comparison

Source: Sunset_Ridge_Forecast.xlsx

Scenario Definitions

Three scenarios were modeled to assess the impact of recurring delays and bottlenecks:

- **Baseline scenario:** current performance continues without intervention
- **Mitigation scenario:** recurring delays are reduced by 50%
- **Worst case scenario:** delays worsen by 25%

Source: Sunset_Ridge_Forecast.xlsx

Scenario Comparison

Scenario	Forecasted Cost	Forecasted Duration
Baseline	232,500	9.8
Mitigation	227,000	8.9
Worst case	240,000	10.5

Source: Sunset_Ridge_Forecast.xlsx

Source: Sunset_Ridge_Forecast.xlsx

Recommendations

- Implement a formal supplier notification process for schedule changes.
- Use weekly progress meetings to coordinate cross-trade dependencies.
- Monitor steel delivery dates closely and escalate early if delays are expected.
- Update the schedule weekly to reflect actual progress and forecast impacts.

Source: Sunset_Ridge_Forecast.xlsx

Scenario Modeling Dashboard

Source: Sunset_Ridge_Forecast.xlsx

09. Risk Register

Objective and Scope

The objective of this section is to identify, assess, and prioritize the risks that could affect the Sunset Ridge Apartments project, using the forecasted cost and schedule performance from Section 07 and the scenario modeling from Section 08. Specifically, it aims to quantify the impact of recurring delays and bottlenecks, compare outcomes across scenarios, and provide a data-driven foundation for decision-making and proactive management.

Key analysis includes:

- Risk identification based on recurring delays and bottlenecks
- Risk assessment using probability and impact scores
- Prioritization of risks by risk score
- Preparation of structured inputs for mitigation planning

Source: Sunset Ridge Apartments Forecasted Cost and Schedule Performance, Weekly Progress Reports, Subcontractor Daily Logs, Inspection Records

Data Sources and Preparation

Primary data sources:

- Risk register: Sunset_Ridge_Risk_Register.xlsx

Data preparation steps:

- Standardized risk IDs and categories
- Calculated risk scores (probability × impact)
- Sorted risks by risk score for prioritization

Source: Sunset_Ridge_Risk_Register.xlsx

Risk Register

Risk ID	Risk	Probability	Impact	Risk Score
R001	Steel delivery delays affecting downstream trades	4	5	20
R002	Crew reassignment without formal communication	3	4	12
R003	Weekly meeting not used for cross-trade coordination	3	4	12
R004	Inspection scheduling not aligned with crew availability	2	3	6

Source: Sunset_Ridge_Risk_Register.xlsx

Source: Sunset_Ridge_Risk_Register.xlsx

Recommendations

- Implement a formal supplier notification process for schedule changes.
- Use weekly progress meetings to coordinate cross-trade dependencies.
- Monitor steel delivery dates closely and escalate early if delays are expected.
- Update the schedule weekly to reflect actual progress and forecast impacts.

Source: Sunset_Ridge_Risk_Register.xlsx

Risk Register Dashboard

Source: Sunset_Ridge_Risk_Register.xlsx

Appendix

Project	Sunset Ridge Apartments
Project Type	Residential Apartment Complex (Wood Frame)
Location	Sunset Ridge, Ontario, Canada
Data Source	Primary data collected from weekly progress reports, subcontractor daily logs, inspection records, and project schedule. Secondary data from baseline budget and schedule.
Analysis Period	March 2025 – May 2025
Tools	Microsoft Excel, Microsoft Word, Microsoft PowerPoint
Prepared by	Project Management Analyst